

Project Title

LeaveApproval Management System Using ServiceNow

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Project Report

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Project Overview

The Smart Leave Approval Management System Using ServiceNow is a workflow automation application developed to simplify and digitize the employee leave approval process within an organization. Instead of relying on manual paperwork, emails, or verbal communication, employees can submit leave requests through a user-friendly ServiceNow form. The system automatically routes the request to the employee's reporting manager for approval using Flow Designer.

Managers can review the request and either approve or reject it. Based on the manager's decision, the leave request status is updated automatically, and email notifications are sent to the employee, ensuring timely communication and transparency throughout the approval process.

The application leverages ServiceNow's low-code capabilities, including custom tables, form design, Flow Designer, approval workflows, email notifications, and role-based access control, to create a secure and efficient leave management solution.

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1. Introduction:

The Smart Leave Approval Management System Using ServiceNow is a workflow automation application designed to simplify and digitize the leave management process within an organization. Traditional leave approval methods often involve paperwork, emails, or manual follow-ups, leading to delays and a lack of transparency.

This project leverages the ServiceNow platform to automate the complete leave approval lifecycle. Employees can submit leave requests through an online form, which are automatically routed to their reporting managers for approval. Managers can approve or reject requests directly within ServiceNow, and employees receive email notifications about the final decision. The system ensures secure data management, faster approvals, and improved operational efficiency.

2. Project Objectives:

The primary objectives of this project are:

- To automate the employee leave request process.
- To eliminate manual paperwork and reduce processing time.
- To route leave requests automatically to the reporting manager.
- To provide a transparent and centralized leave approval system.
- To notify employees automatically when their leave is approved or rejected.
- To maintain a complete history of leave requests for future reference.
- To demonstrate ServiceNow application development using App Engine Studio and Flow Designer.

3. Key Features

The Leave Approval Management System provides the following features:

- Employee Leave Request Form
- Automatic Manager Approval Workflow
- Flow Designer-based Approval Process
- Email Notifications for Approval and Rejection
- Custom Leave Request Table
- Role-Based Access Control
- Leave Status Tracking (Pending, Approved, Rejected)
- Date Validation for Leave Requests
- Manager Dashboard for Approvals
- Secure and Centralized Data Storage

4. Project Milestones

The project implementation follows five major milestones.

Milestone 1: Application Development

Objective

Develop a custom ServiceNow application to manage employee leave requests and approvals.

Activities

- Create a custom scoped application.
- Create the **Leave Request** table.
- Configure application modules.
- Design the Leave Request form.
- Add required fields such as Employee, Manager, Leave Type, Start Date, End Date, and Status.
- Organize the application according to ServiceNow development best practices.

Milestone 2: Form Configuration & Validation

Objective

Design an interactive leave request form and ensure accurate data collection.

- Configure form layout and form design.
- Create choice and reference fields.

Activities

- Implement Client Script for date validation.
- Prevent End Date from being earlier than Start Date.
- Configure mandatory fields.
- Improve user experience through form customization.

Milestone 3: Workflow & Approval Integration

Objective

Automate the leave approval process using ServiceNow Flow Designer.

- Create Flow Designer workflow.
- Configure Record Created trigger.

Activities

- Implement Manager Approval using **Ask For Approval** action.
- Update leave request status after approval or rejection.
- Configure email notifications.
- Maintain approval records for auditing purposes.

Milestone 4: Testing

Objective

Validate the functionality and performance of the Leave Approval Management System.

- Unit testing of individual components.
- Leave request form testing.

Activities

- Client Script validation testing.
- Flow Designer workflow testing.
- Manager approval and rejection testing.
- Email notification testing.
- End-to-end workflow testing.
- User Acceptance Testing (UAT).

Milestone 5: Deployment & Documentation

Objective

Deploy the application and prepare complete project documentation.

- Verify application configuration.
- Publish the Flow Designer workflow.

Activities

- Validate application functionality.
- Prepare project documentation.
- Capture implementation screenshots.
- Demonstrate the application.
- Document future enhancements and recommendations.
- Final project submission.

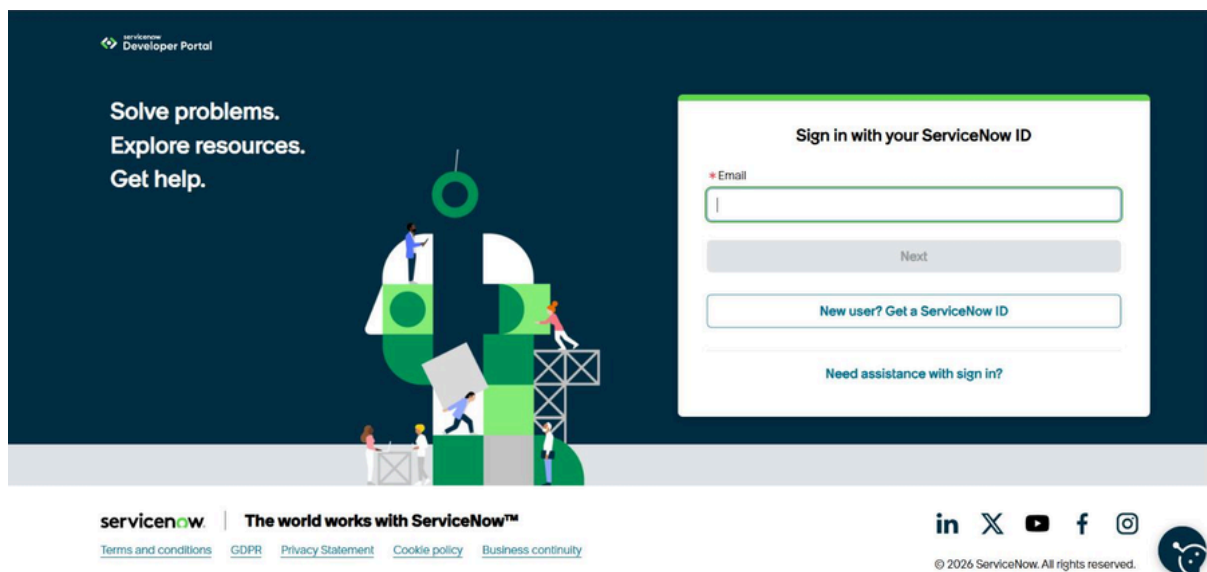
5. ServiceNow Developer Setup

Before developing the application, a ServiceNow Personal Developer Instance (PDI) must be created. The PDI provides a fully functional ServiceNow environment where applications can be developed, tested, and deployed without affecting production systems.

5.1 Creating a ServiceNow Developer Account :

To begin development, perform the following steps:

1. Visit the ServiceNow Developer Portal.
2. Click **Sign Up** to create a free developer account.
3. Enter the required information:
 - First Name
 - Last Name
 - Email Address
 - Password
4. Verify the email address using the activation link sent to your registered email.
5. Log in to the ServiceNow Developer Portal.
6. Click **Start Building**.
7. Request a **Personal Developer Instance (PDI)**.
8. Wait for the instance to be provisioned.
9. Launch the instance and log in using the generated credentials.



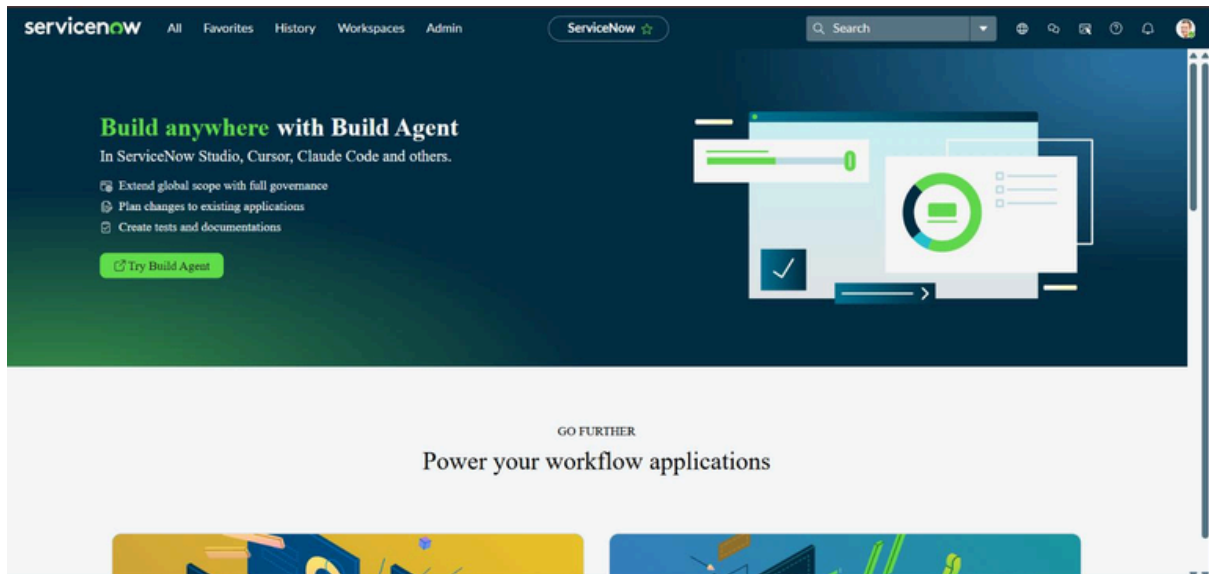
5.2 Accessing Creator Studio

Once the Personal Developer Instance is ready, ServiceNow redirects the developer to Creator Studio.

Creator Studio provides a no-code and low-code environment for building applications. It simplifies application development by allowing developers to create forms, workflows, tables, automation, and integrations through an intuitive graphical interface.

Creator Studio offers features such as:

- Application Creation
- Data Modeling
- Flow Automation
- Form Builder
- Catalog Item Development
- Workflow Configuration



6. Project Implementation in ServiceNow

6.1 ApplicationCreation

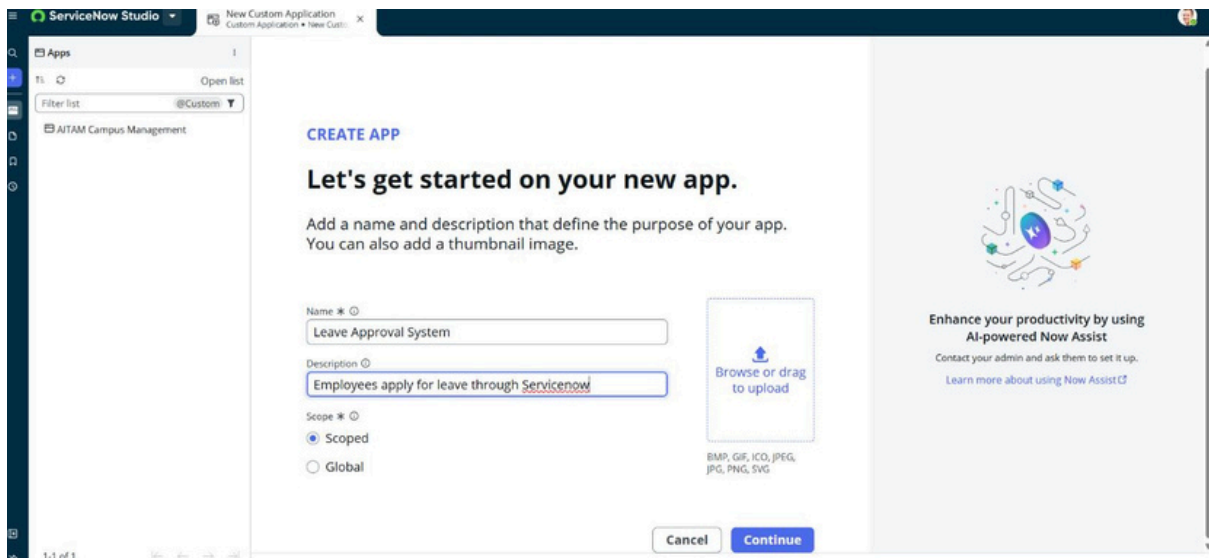
The Leave Approval Management System was developed as a custom scoped application in ServiceNow using **App Engine Studio**. The application provides an automated workflow for employees to submit leave requests and managers to review and approve or reject them. Developing the application within a scoped environment ensures security, modularity, and ease of maintenance.

Steps to Create the Application

1. Open the **Application Navigator** in the ServiceNow platform.
2. Search for **App Engine Studio**.
3. Click **Create App**.
4. Select **Build an App from Scratch**.
5. Enter the following details:
 - o **Application Name:** Leave Approval Management System
 - o **Description:** Automates employee leave request and approval process.
6. Click **Create** to generate the scoped application.

Purpose of the Application

The application provides a centralized platform where employees can submit leave requests and managers can review and approve them through an automated workflow. It replaces manual leave management processes with a digital solution, improving efficiency, transparency, and record management.



6.2 Leave Request Table Creation

A custom table named **Leave Request** was created to store all employee leave records. This table serves as the primary database for maintaining leave information, approval status, and request history.

Steps to Create the Leave Request Table

1. Open **App Engine Studio**.
2. Navigate to the **Data** section.
3. Click **Create Table**.
4. Enter the following details:
 - o **Table Label:** Leave Request
 - o **Table Name:** Leave Request

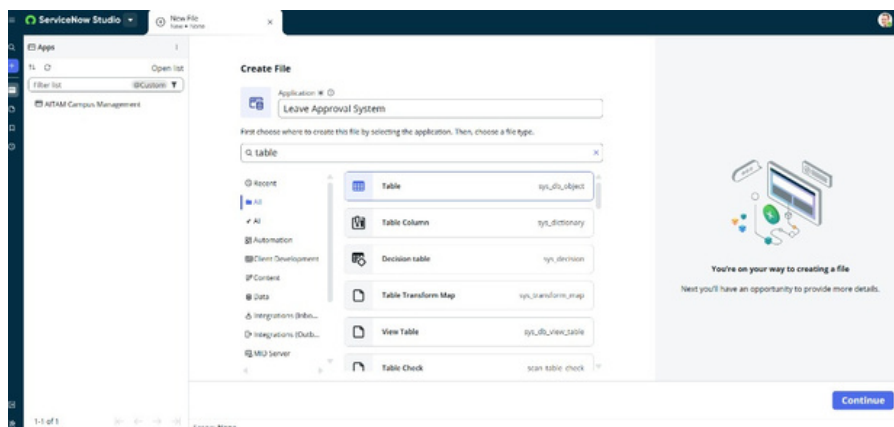
5. Save the table.

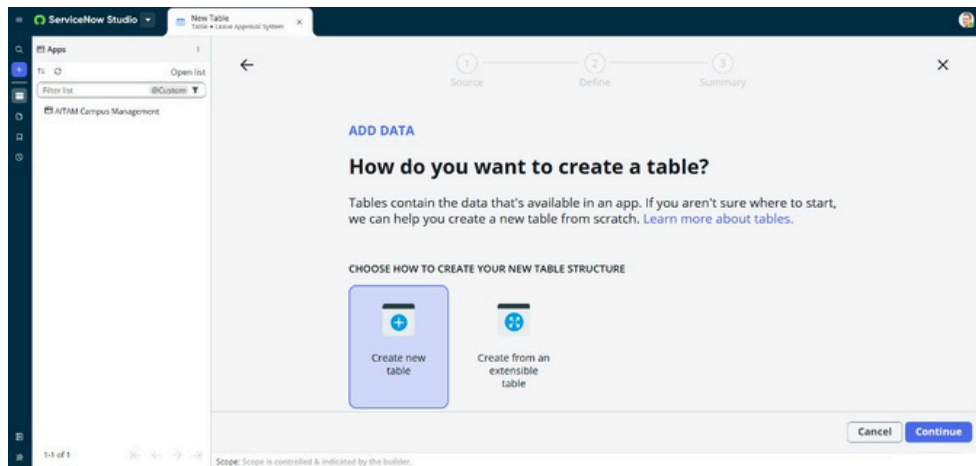
Purpose of the Table

The Leave Request table stores all information related to employee leave requests and enables managers to review, approve, or reject requests efficiently.

Advantages

- Centralized storage of leave records.
- Easy tracking of approval status.
- Simplified reporting and auditing.
- Supports workflow automation.





6.3 Field Creation

After creating the custom table, the required fields were added to capture leave request information.

Steps to Create Fields

1. Open the **Leave Request** table.
2. Navigate to the **Columns** section.
3. Click **New**.
4. Select the appropriate field type.
5. Enter the field label and column name.
6. Save the field.
7. Repeat the process for all required fields.

Fields Created

Field Name Data Type

Employee Reference (User)

Manager Reference (User)

Leave Type Choice

Start Date Date

End Date Date

Status Choice

Reason String

6.4 Form Design

The Leave Request form was designed to allow employees to submit leave requests easily. Only the required fields are displayed to ensure a simple and user-friendly interface.

Objective

- Collect complete leave information.
- Improve user experience.
- Reduce data entry errors.
- Simplify leave request submission.

Form Components

- Employee
 - Manager
 - Leave Type
 - Start Date
 - End Date
 - Reason
 - Status
-

6.5 Client Script Implementation

A Client Script was implemented to validate the selected leave dates before submission.

Objective

Prevent employees from selecting an End Date earlier than the Start Date.

Steps

1. Navigate to **System Definition → Client Scripts**.
2. Create a new Client Script.
3. Select the **Leave Request** table.
4. Configure the script to validate the dates.
5. Save the Client Script.

Purpose

- Prevent invalid leave requests.
 - Improve data accuracy.
 - Enhance user experience.
-

6.6 Flow Designer Implementation

Flow Designer automates the complete leave approval process. Whenever a new leave request is created, the flow automatically routes it to the manager for approval.

6.6.1 Flow Creation

Steps

1. Open **Flow Designer** .
 2. Click **New → Flow** .
 3. Enter:
 - o **Flow Name:** Leave Approval Workflow
 - o **Description:** Automates leave approval requests.
 4. Click **Build Flow**.
-

6.6.2 Configuring the Trigger

The workflow starts automatically whenever a new Leave Request record is created.

- Record Created

- Table: Leave Request

6.6.3 ManagerApproval

The **Ask for Approval** action sends the leave request to the selected manager.

Steps

1. Add **Ask for Approval** action.
2. Select the Leave Request record.
3. Configure:
 - o Approver = Manager
 - o Approval Rule = Anyone Approves
4. Save the action.

Approval Outcomes

- Approved
- Rejected
- Pending

6.6.4Flow Logic

After the manager submits a decision, the workflow automatically determines the next action.

Approved

- Update Status to **Approved**.
- Send approval email.
- Complete the workflow.

Rejected

- Update Status to **Rejected**.
- Send rejection email.
- Stop further processing.

6.6.5 Email Notifications

The system automatically sends notifications to employees during important stages.

Notifications

- Leave Request Submitted
- Leave Approved
- Leave Rejected

Benefits

- Keeps employees informed.
- Reduces manual communication.
- Improves transparency.

6.6.6 Update Record

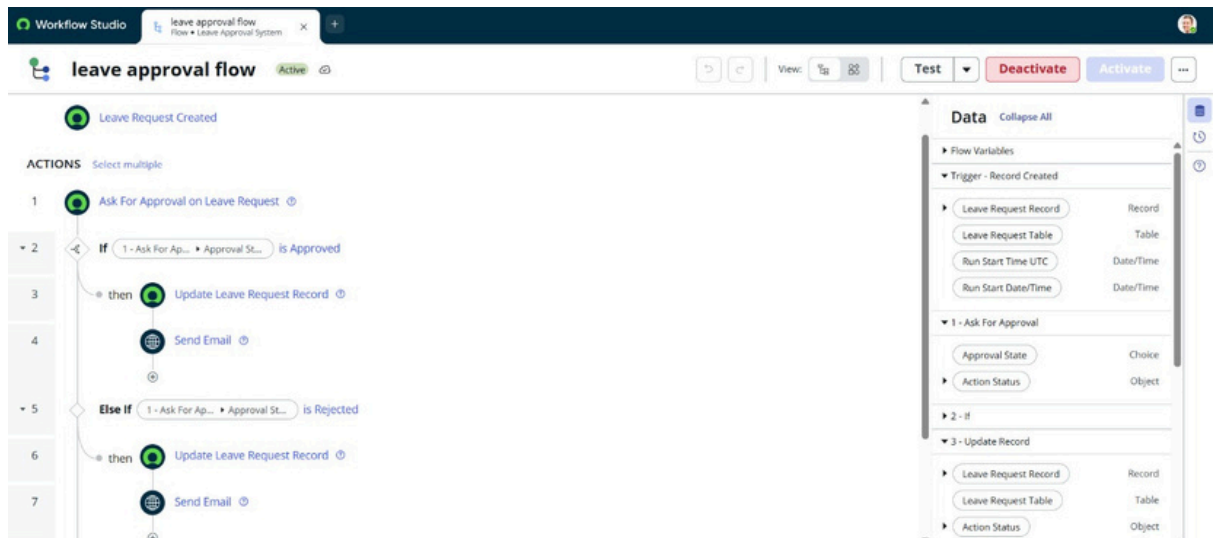
After the approval process is completed, the Leave Request record is updated automatically.

Updated Fields

- Status
- Approval Date
- Manager Decision

Benefits

- Maintains accurate leave records.
- Simplifies reporting.
- Tracks approval history.



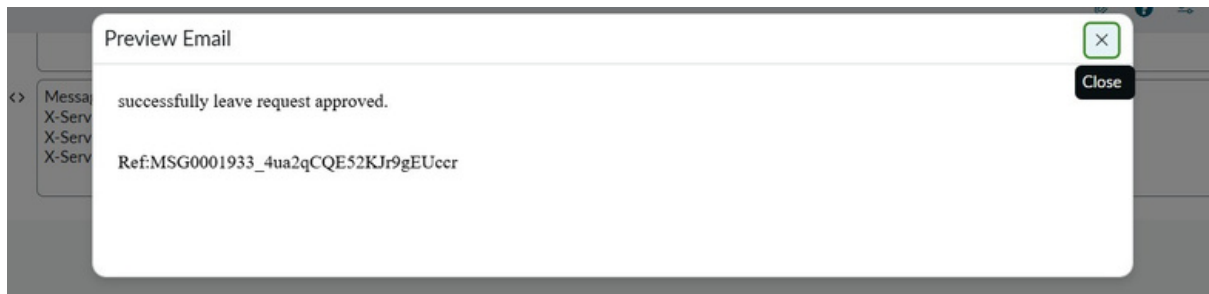
6.7 Workflow Summary

The Leave Approval Management System follows the workflow below:

1. Employee submits a Leave Request.
2. Flow Designer is triggered automatically.
3. Leave request is routed to the Manager.
4. Manager reviews the request.
5. Manager approves or rejects the request.
6. Flow updates the leave request status.
7. Email notification is sent to the employee.
8. The workflow is completed, and the leave request is stored for future reference.

7. Output Screenshots

The screenshot shows the ServiceNow 'Approval - Leave Request: LEA1030' form. The form has a header with 'Approval - Leave Request: LEA1030' and 'Application scope: Leave Approval System'. The form fields include 'Approver' (David Dan), 'State' (Approved), 'Approval Reason', and 'Comments'. The 'Activities' section shows two activities: 'System Administrator' with 'State: Approved was Requested' and 'System Administrator' with 'Approver: David Dan, State: Requested'. The form has 'Update' and 'Delete' buttons at the bottom.



8. Conclusion

The **Leave Approval Management System** developed using the ServiceNow platform demonstrates how workflow automation can simplify and streamline the employee leave management process. By utilizing ServiceNow features such as **App Engine Studio, Custom Tables, Form Design, Flow Designer, Client Scripts, Approval Management, and Email Notifications**, the application provides an efficient and user-friendly solution for managing leave requests from submission to final approval.

The automated workflow eliminates the need for manual paperwork and repetitive follow-ups by routing leave requests directly to the appropriate manager for approval. Automated status updates and email notifications keep employees informed throughout the approval process, improving communication, transparency, and accountability within the organization.

The project also ensures data accuracy by validating leave dates before submission and maintaining a centralized repository of all leave requests for easy tracking, reporting, and auditing. This reduces administrative effort, minimizes processing delays, and provides a standardized leave approval process.

In conclusion, the **Leave Approval Management System** is a practical and scalable solution that enhances organizational efficiency through automation. The project demonstrates the capabilities of the ServiceNow platform in developing real-world business applications and provides a strong foundation for future enhancements such as leave balance management, multi-level approvals, HR integration, reporting dashboards, and mobile accessibility.